

FoodBridge - Product Requirements Document (PRD)

Product Vision

Enable restaurants, hotels, event organizers, NGOs, and volunteers to collaborate through a centralized platform that reduces food waste and ensures surplus food reaches people in need efficiently and safely.

Problem Statement

Large quantities of edible food are wasted every day by restaurants, hotels, cafeterias, and event organizers. At the same time, many individuals and communities experience food insecurity. Current food donation processes are fragmented, manual, and lack transparency, resulting in delayed distribution and unnecessary food waste.

Product Goals :

Goal ID	Goal	KPI
PG-01	Reduce food waste	Reduce reported food wastage by 40%
PG-02	Improve donation efficiency	Average donation processing time < 2 hours
PG-03	Increase successful deliveries	90% of accepted donations delivered successfully
PG-04	Build stakeholder trust	User satisfaction score $\geq$ 85%

SMART Requirement Writing Standard

All requirements are written following SMART principles.

SMART Element	Application in FoodBridge
Specific	Requirements clearly define donor, NGO, volunteer, and admin actions
Measurable	Success is measured using KPIs and delivery statistics
Achievable	Scope is limited to food donation and distribution management
Relevant	Directly addresses food waste and food insecurity
Timely	Requirements are organized by project milestones and releases

MoSCoW Prioritization :

Category	Definition	FoodBridge Usage
Must Have	Essential features	Authentication, Donation Posting, Food Requests, Delivery Tracking
Should Have	Important enhancements	Notifications, Search & Filter, Reports
Could Have	Optional improvements	Analytics Dashboard, Donation History Insights
Won't Have	Out of scope for current release	AI Food Prediction, Government Integration

Target Users:

- Restaurants
- Hotels
- Event Organizers
- NGOs
- Volunteers
- Administrators

FoodBridge - User Personas:

Persona P1 - Ahmed Hassan (Restaurant Owner)	
Attribute	Details
Demographics	42, owner of a medium-sized restaurant in Dhaka
Goals	Donate surplus food before expiration and reduce food waste
Motivations	Social responsibility, environmental sustainability, community support
Pain Points	Difficulty finding NGOs quickly, food spoilage before collection
Technology Usage	Smartphone and web application user
Typical Workflow	Identify surplus food → Create donation post → Wait for NGO request → Volunteer pickup → Track delivery

Persona P2 - Fatema Akter (NGO Coordinator)	
Attribute	Details
Demographics	35, NGO coordinator managing food distribution programs
Goals	Find food donations quickly and distribute them to beneficiaries
Motivations	Support vulnerable communities and reduce hunger
Pain Points	Limited visibility of available donations, delayed communication
Technology Usage	Laptop and mobile user, experienced with management systems

Persona P2 - Fatema Akter (NGO Coordinator)

Typical Workflow	Browse donations → Request food → Track approval → Receive delivery → Confirm receipt
------------------	---

Persona P3 - Hasan Rahman (Volunteer)

Attribute	Details
Demographics	24, university student and community volunteer
Goals	Participate in social activities and assist food delivery
Motivations	Community service and personal contribution to society
Pain Points	Lack of organized delivery assignments and communication issues
Technology Usage	Mobile-first user, frequent use of maps and navigation apps
Typical Workflow	Login → View delivery requests → Accept task → Pickup food → Deliver food → Update status

Persona P4 - System Administrator

Attribute	Details
Demographics	30, platform administrator
Goals	Maintain system security, monitor operations, and manage users

Persona P4 - System Administrator

Motivations	Ensure platform reliability and stakeholder satisfaction
Pain Points	Managing fraudulent accounts, monitoring large datasets
Technology Usage	Desktop-first user with advanced system access
Typical Workflow	Review users → Verify NGOs → Monitor donations → Generate reports → Resolve issues

Persona P5 -Event Organizer

Attribute	Details
Demographics	38, event management professional
Goals	Donate leftover food from events efficiently
Motivations	Reduce food waste and improve organizational reputation
Pain Points	Large volumes of surplus food and limited donation channels
Technology Usage	Mobile and desktop web user
Typical Workflow	Complete event → Identify leftover food → Post donation → Coordinate pickup → Confirm donation completion

FoodBridge - User Journey:

The user journey of FoodBridge begins when a user visits the platform and creates an account by completing the registration process. After successful registration, the user logs into the system using valid credentials and gains access to the dashboard based on their role. A food donor, such as a restaurant owner or event organizer, can then create a donation by providing details about the available food, including quantity, type, expiration time, and pickup location. Once the donation is published, it becomes visible to registered NGOs on the platform.

NGO representatives can browse available food donations and submit requests for the donations that match their requirements. After receiving a request, the donor can review and approve it. Once approved, the system assigns or allows a volunteer to accept the delivery task. The volunteer receives the pickup and delivery information and proceeds to collect the food from the donor's location.

After collecting the food, the volunteer transports it to the requesting NGO or designated distribution center. Throughout the process, the system updates the donation status and notifies all stakeholders about important events, such as request approval, pickup confirmation, and delivery progress. When the food is successfully delivered, the NGO confirms receipt through the platform. The donation is then marked as completed, and the transaction is stored in the system records for future tracking, reporting, and transparency purposes.

This user journey ensures efficient coordination among donors, NGOs, volunteers, and administrators, helping to reduce food waste and improve food distribution to communities in need.

FoodBridge - User Stories:

Epic E1 - User Authentication and Account Management

As a donor, NGO representative, volunteer, or administrator, I want to register an account and securely log in to the platform so that I can access features based on my role. I also want to update my profile information and recover my account if I forget my password.

Epic E2 - Food Donation Management

As a food donor, I want to create, edit, and manage food donation listings so that surplus food can be distributed instead of wasted. I also want to track the status of my donations and receive notifications when an NGO requests food or when a delivery is completed.

Epic E3 - Food Request Management

As an NGO representative, I want to browse available food donations and submit requests for suitable donations so that I can collect food for people in need. I also want to track the progress of my requests and receive updates regarding approval and delivery.

Epic E4 - Volunteer Management

As a volunteer, I want to view available delivery assignments and accept tasks that match my availability so that I can help transport food from donors to NGOs. I also want to update the delivery status after pickup and delivery.

Epic E5 - Notification and Tracking

As a system user, I want to receive notifications about important activities such as donation requests, approvals, volunteer assignments, and delivery updates so that I can stay informed throughout the donation process.

Epic E6 - Administration and Monitoring

As an administrator, I want to manage users, verify NGOs, monitor donations and deliveries, and generate reports so that I can ensure the platform operates efficiently, securely, and transparently.

FoodBridge - Acceptance Criteria:

User Registration

The system shall allow new users, including donors, NGO representatives, volunteers, and administrators, to create an account by providing valid registration information. Upon successful registration, the system shall store the user information securely and provide access to the platform according to the assigned role.

User Authentication

The system shall allow registered users to log in using valid credentials. Upon successful authentication, users shall be redirected to their respective dashboards. Invalid login attempts shall generate appropriate error messages.

Food Donation Creation

The system shall allow donors to create food donation listings by providing food details such as food type, quantity, expiration time, and pickup location. Once submitted, the donation shall be stored in the database and displayed to eligible NGOs.

Food Request Submission

The system shall allow NGO representatives to browse available food donations and submit requests for desired donations. The submitted request shall be recorded and made visible to the corresponding donor for review and approval.

Donation Approval

The system shall allow donors to review incoming requests and approve or reject them. Once approved, the system shall notify the requesting NGO and make the donation available for delivery assignment.

Volunteer Assignment

The system shall allow volunteers to view available delivery tasks and accept assignments. Upon acceptance, the system shall provide pickup and delivery information and update the task status accordingly.

Food Delivery Tracking

The system shall track the progress of food delivery from pickup to final destination. All stakeholders shall receive status updates regarding the delivery process.

Delivery Confirmation

The system shall allow NGOs to confirm receipt of donated food after successful delivery. Once confirmation is received, the donation status shall be updated to "Completed."

Notification Management

The system shall send notifications for important events, including donation requests, approvals, volunteer assignments, pickup confirmations, and delivery completions. Notifications shall be delivered in a timely manner to the relevant stakeholders.

Administrative Management

The system shall allow administrators to manage users, verify NGO accounts, monitor donation activities, and generate reports. Administrators shall have access to system monitoring tools and management functionalities necessary for maintaining platform integrity.

Data Security and Reliability

The system shall protect user information through secure authentication and authorization mechanisms. All donation records, requests, and delivery information shall be stored reliably and remain accessible to authorized users only.